

Incorporating Process Information Into Cognitive Diagnostic Models: A Four-Component Joint Modeling Approach

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Recent studies show increasing interest in using process data (e.g., response time, response actions) to enhance measurement accuracy for respondents' latent traits. Yet, few have explored the possibility of incorporating process information into cognitive diagnostic models (CDMs). This study proposes a novel CDM approach that utilizes a four-component joint modeling approach with response action sequences (i.e., similarity and efficiency), response time, and item responses. We employed the Markov Chain Monte Carlo method for parameter estimation and evaluated the performance of the proposed model using both an empirical study and two simulation studies. The results suggest that the process data can improve respondents' classification accuracy under varied conditions and support the interpretation of the association between process and response data.

Keywords: *process data; log file; action sequence; cognitive diagnostic model; response time*

Cognitive diagnosis in educational contexts serves to evaluate the learning process by providing diagnostic information (Kaya & Leite, 2017). Cognitive diagnostic models (CDMs) aim to measure the mastery of specific skills or attributes based on examinees' responses to test items (Kaya & Leite, 2017; Ma & Guo, 2019). CDMs are crucial in cognitive assessment as they provide nuanced information about examinees' skill profiles based on underlying mental

components in the response process, thereby enabling the facilitation of targeted interventions for cognitive improvement (Rupp & Templin, 2007). While item response data have been widely used for inferring examinees' attributes using CDMs, other information may be considered when it is available to improve the estimation accuracy (Bezirhan, 2021; De Boeck & Jeon, 2019; van der Linden, 2007, 2009; Zhan et al., 2018).

With the increased adoption of computer-based testing and e-learning, process data is usually captured in log files, offering not only insights into individuals' responses to tasks but also their problem-solving approaches (Goldhammer et al., 2013; He et al., 2021). Process data is significantly associated with individuals' proficiencies (Liang et al., 2023), making it an invaluable source of information for refining educational measurements. Its role evolved to be a source of supplementary information to improve proficiency estimation, measurement accuracy, and test designs (He, Borgonovi, & Paccagnella, 2019). Consequently, process data aids in exploring the intrinsic psychometric approaches behind test performances, a level of understanding that may not be possible to achieve by relying solely on response data (Goldhammer et al., 2013).

In recent literature, most researchers have utilized response time (RT) as an additional source of information to improve the classification and estimation accuracy of CDMs (see Bolsinova, de Boeck, & Tijmstra, 2017; Fox & Marianti, 2016; Klein Entink et al., 2009; Man et al., 2019; Zhan et al., 2018). RT is the time an examinee takes to complete the cognitive process to answer a certain item in a test (Lee & Chen, 2011). The effect of RT on response accuracy was widely studied (see Klein et al., 2009; van der Linden, 2007, 2009), and many found positive inter-dependencies between RT and response accuracy (see Klein et al., 2009; Roberts & Stankov, 1999; van der Linden & Fox, 2016), indicating test takers with high abilities tend to respond slowly, especially in items with high difficulty (Domingue et al., 2022). Such trade-offs between RT and response accuracy are simultaneous within person-characteristics for the same assessment items (Bezirhan, 2021). Prior research indicates that RTs can be augmented within current psychometric models as a supplementary covariate to enhance the precision of both ability and item parameters estimation (Liang et al., 2023). Beyond RT, other studies considered process data such as eye-tracking data (Bergner & von Davier, 2019; Man & Harring, 2021), mouse-click or drag data (Liang et al., 2023), writing keystrokes (Guo et al., 2018), and action sequence data (He, Liao, & Jiao, 2019; He & von Davier, 2016).

Action sequence usually indicates a series of actions or steps undertaken by test-takers while solving a problem (Tang et al., 2021). This can encompass a range of behaviors, including mouse clicks, keyboard entries, time spent on different parts of a question, navigation patterns through the test, or any other recorded interactions with the test interface. Action sequence provides

supplementary information on the latent ability of test-takers (LaMar, 2018) and is significantly associated with test performance (He & von Davier, 2015, 2016). Employing a sequence mining technique, He, Borgonovi, and Paccagnella (2019) demonstrated a promising way to summarize test takers' action sequences using two measures (i.e., response action similarity and response action efficiency). Specifically, action similarity and action efficiency were developed based on the longest common subsequence (LCS), which measures the distances between the action sequence of a test-taker to a specific test item and the action sequence deemed optimal by content experts for solving the item (He, Borgonovi, & Paccagnella, 2019). Response action similarity quantifies the proportion of the common action subsequence (performed by the test taker) present in the reference action sequence, and response action efficiency quantifies the extent to which the LCS and the individual action sequence match. Integrating information from the two action sequence measures would enhance the precision of ability estimation in cognitive assessments (He, Liao, & Jiao, 2019; He et al., 2023).

Despite current progress in incorporating process data into CDMs, there is still a scarcity of research that augments multiple types of process data to refine ability estimations (Liang et al., 2023). In addition, compared to the existing literature, this study adopts a unique way to summarize process data relevant to action sequence and integrates it into the CDM framework. This study presents an extended hierarchical cognitive assessment model, that is, the four-component CDM, by incorporating response accuracy, RT, response action similarity, and response action efficiency for improved ability or proficiency estimations. Among these three process-related variables, RT is commonly used in joint models to consider the interrelationship between response time and response accuracy, while the action similarity and efficiency are general measures for action sequence after the assessment was processed, which provide possibilities to track back the respondents' problem-solving procedures.

In the subsequent sections, we first propose the four-component CDM and explain the process of model estimation employing a Bayesian approach. Next, we demonstrate an empirical application with the use of the proposed four-component model on response data and process data collected in the problem solving in technology rich environments (PSTRE) subject in the Program for the International Assessment of Adult Competencies (PIAAC) 2012 survey data (OECD, 2012). Furthermore, we compared the empirical performance of the four-component model with existing models. In the following sections, we present two simulation studies. The first simulation study was conducted with the estimated parameters from the empirical study, with a focus on evaluating the model performance when incorporating RT, response action similarity, and efficiency within the CDM framework. The second simulation study was conducted

under varying simulation conditions to explore the performance of the four-component model accordingly.

The Proposed CDM

This study adopts the hierarchical joint modeling framework (van der Linden, 2007) to integrate three additional process data components into CDM to enhance estimation accuracy. Previous studies have used a similar hierarchical joint modeling approach to integrate process data into CDM frameworks (see more in: Fox & Marianti, 2016; Liang et al., 2023; Man et al., 2019; van der Linden, 2007; Zhan et al., 2018). Among process data variables integrated into the four-component model, RT can be directly collected from the log file, while similarity and efficiency variables have to be computed based on test takers' action sequences (see detailed definition and computation in the following section). It is important to note that both similarity and efficiency variables are indirect measures based on test-takers' action sequences. Action sequences are the records of all human-computer interactions, such as mouse clicks and keystrokes, during the whole process of an individual solving a digital problem. It was found that the strategies taken in the problem-solving process have an association with test takers' latent abilities (He, Borgonovi, & Paccagnella, 2019; LaMar, 2018). However, despite the promise of action sequences to contribute to the measurement, this data source has not been commonly used yet, because of the challenges in handling their high dimensionality and complexity of interpretation of their sequential property. He, Borgonovi, and Paccagnella (2019) utilized the LCS method to summarize the information extracted from action sequences and proposed action similarity and action efficiency measures that "can reliably capture and characterize the strategies followed by respondents across a variety of problem-solving assessment items" (p. 40). Considering the significant analytical potential that both action similarity and efficiency measures can contribute to the cognitive assessment, we propose a four-component CDM to accommodate supplementary information from these two measures (along with RT) to enhance cognitive assessment. The proposed model utilized the variance-covariance structure of persons' parameters from all process information sources to set up a hierarchical joint modeling framework. Specifically, the four-component model consists of four separate measurement models for response accuracy, response time, response action similarity, and response action efficiency, and a structural model that accounts for the associations among person parameters. The hierarchical joint modeling framework of the four-component model is presented in Figure 1, and more details are given below.

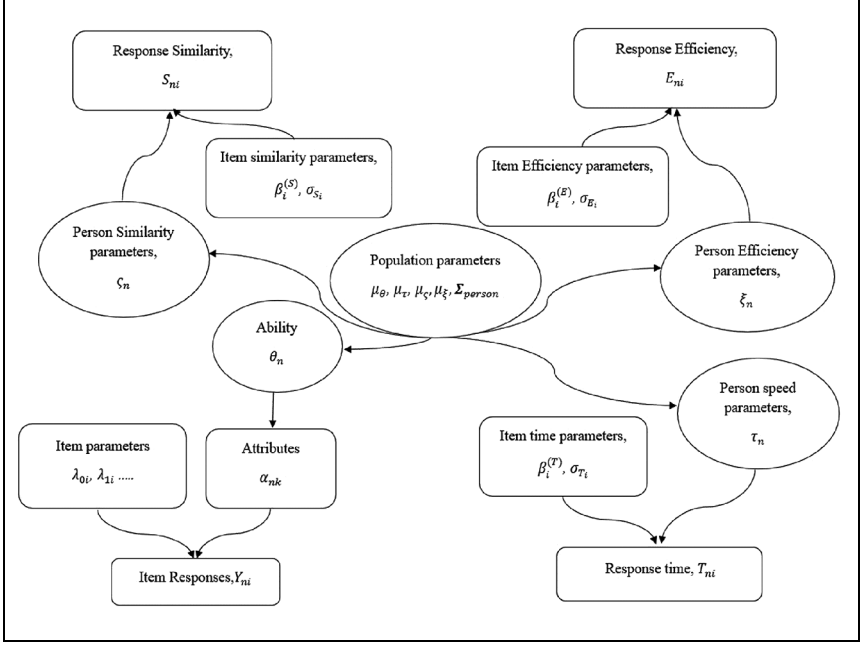


FIGURE 1. *The four-component model: Under a theoretical framework.*

Measurement Models

Response Accuracy. Let, Y_{ni} denotes the response to item i by person n , where $Y_{ni} = 1$ when the response is correct and $Y_{ni} = 0$, otherwise. Assume a test measures K ($k = 1, \dots, K$) binary attributes (i.e., mastery vs. non-mastery) with I ($i = 1, \dots, I$) items. As the attributes are assumed to be binary, there are 2^K latent classes. Suppose the first K_i^* denote the number of attributes measured by the item i , and the reduced attribute vector for the item i for person n is α_{nli}^* , where l represents the latent group for the item i . If an item measures K_i^* attributes, then there are $2^{K_i^*}$ latent groups (i.e., $l = 1, \dots, 2^{K_i^*}$). As a widely applied model in cognitive diagnostic assessments, the linear logistic model (LLM; Maris, 1999) is adopted as the measurement model for response accuracy. The item response function of the LLM is presented as

$$\text{logit}[P(Y_{ni} = 1 | \alpha_{nli}^*)] = \lambda_{i0} + \sum_{k=1}^{K_i^*} \lambda_{ik} \alpha_{nik}, \quad (1)$$

where $P(Y_{ni} = 1 | \alpha_{nli}^*)$ is the probability that person n answers item i correctly, given the examinee's latent attribute vector α_{nli}^* , λ_{i0} is the intercept of item i ,

and λ_{ik} is the parameter representing the impact on the log-odds of a correct response to item i if attribute k is mastered. LLM is widely applied in cognitive diagnostic assessments.

Response Time. RT is commonly modeled with parametric distributions (Bezirhan, 2021; Klein et al., 2009). Appropriate specification of RT distribution is a crucial issue in RT modeling (Van Zandt, 2002). Raw RT data are often transformed so that the residuals from the RT model are more normally distributed (Balota et al., 2013). Previous studies indicated that the lognormal RT model performed excellently, given that RT is almost always positively skewed (see van der Linden, 2007; van der Linden & Guo, 2008; Zhan et al., 2018). This study adopts the lognormal RT model in the four-component model, as in

$$\log(T_{ni}) = \tau_n - \beta_i^{(T)} + \varepsilon_{ni}; \quad \varepsilon_{ni} \sim N(0, \sigma_{Ti}^2), \quad (2)$$

where T_{ni} is the response time to answer item i for person n , τ_n is the person's speed parameter, $\beta_i^{(T)}$ is the population time parameter for item i , and ε_{ni} is the normally distributed error term. In this lognormal RT model, the person speed parameter τ_n can be interpreted as the mean log (speed) of person n for responding to a certain item, and the parameter $\beta_i^{(T)}$ can be identified as the average log (response time) of item i for all respondents. To interpret $\beta_i^{(T)}$, we must account for the negative sign embedded with $\beta_i^{(T)}$ in Equation 2, that is, $(-\beta_i^{(T)})$ can be interpreted as the average log (response time) of item i for all respondents.

Response Action Similarity. Response action similarity (named similarity hereafter) is an indicator to measure the degree of similarity between the individual's observed sequence and predefined reference sequence (i.e., optimal action sequence defined by the content experts and item developers) by calculating the sequence distance via the LCS (more in: He, Borgonovi, & Paccagnella, 2019). For each item, the similarity is defined as the ratio between the length of the LCS and the length of the reference sequence with the aim of detecting the degree of similar actions an individual used compared with the optimal reference sequence. Similarity can be calculated as (He, Borgonovi, & Paccagnella, 2019)

$$\text{Similarity} = \frac{\text{Length (LCS)}}{\text{Length (Reference Sequence)}}.$$

Since the length of LCS cannot be greater than the length of the reference sequence, the similarity value will always be within the range of zero to one. A similarity value of zero indicates no match between the LCS and the optimum

reference sequence, and a similarity value of one indicates a perfect match between the LCS and the optimum reference sequence. A higher similarity value indicates a greater match between the individual sequence and the reference sequence.

It is noted that multiple optimal solutions are often designed for items. For items with multiple predefined reference sequences, we employed similar methods in He, Borgonovi, and Paccagnella (2019) to compute the LCS between individual observed sequences and each predefined reference sequence and selected the longest length of LCS as the most likely strategy that the respondent took. The similarity score for this item was then computed based on the most likely strategy (i.e., reference sequence). Specifically, it is the ratio between the longest length of LCS and the length of the specific corresponding reference sequence that the respondent most likely employed the strategy.

Let S_{ni} be the similarity score for person n and item i . As S_{ni} take values between zero and one. We define

$$\text{logit}(S_{ni}) = s_n - \beta_i^{(S)} + v_{ni}; v_{ni} \sim N(0, \sigma_{Si}^2), \quad (3)$$

where s_n is the persons' similarity parameter, $\beta_i^{(S)}$ is the population similarity parameter, and v_{ni} is the normally distributed error term. Person similarity parameter s_n can be interpreted as the mean of logit (similarity) for person n for certain items, and the parameter $\beta_i^{(S)}$ can be identified as the average logit (similarity) for responding to item i for all respondents. A higher value of $\beta_i^{(S)}$ indicates that all respondents, on average, perform more similar actions (compared to the optimal sequence for item i). To interpret $\beta_i^{(S)}$, we need to consider the negative sign embedded with $\beta_i^{(S)}$ in Equation 3.

Response Action Efficiency. Response action efficiency (named efficiency hereafter) is another indicator that summarizes the degree of match between test takers' LCS and observed sequence (i.e., a sequence performed by a test taker while responding to an item) with the aim of detecting the degree of redundant actions used. To derive response action efficiency, test-takers' LCS is compared with the respective observed sequence for specific items, which is calculated as (He, Borgonovi, & Paccagnella, 2019)

$$\text{Efficiency} = \frac{\text{Length (LCS)}}{\text{Length (Observed Sequence)}}.$$

The length of LCS is always less than or equal to the length of the observed sequence. Hence, the efficiency value for any individual for any specific item would always be between zero and one. An efficiency value close to one indicates that the test-taker responded efficiently to the item, avoiding excessive actions that are not part of the optimal sequence.

Analogously to the computation of a similarity score, in the situation of multiple optimal reference sequences, the efficiency score for such an item was computed based on the most likely strategy (i.e., reference sequence). Specifically, it is the ratio between the longest length of LCS that the respondent most likely employed the strategy and the length of the individual observed sequence.

Let, E_{ni} be the efficiency score for person n , and item i . As E_{ni} take values between zero and one, we define,

$$\text{logit}(E_{ni}) = \xi_n - \beta_i^{(E)} + \omega_{ni}; \quad \omega_{ni} \sim N(0, \sigma_{Ei}^2), \quad (4)$$

where ξ_n is the persons' efficiency parameter, $\beta_i^{(E)}$ is the population efficiency parameter, and ω_{ni} is the normally distributed error term. Person efficiency parameter ξ_n can be interpreted as the mean $\text{logit}(\text{efficiency})$ for person n for a specific item, and the parameter $\beta_i^{(E)}$ can be identified as the average $\text{logit}(\text{efficiency})$ for responding to item i for all respondents. A higher value of $\beta_i^{(E)}$ indicates that all respondents, on average, perform more efficient actions (compared to the optimal sequence for item i). To interpret $\beta_i^{(E)}$, we need to consider the negative sign embedded with $\beta_i^{(E)}$ in Equation 4.

Joint Distribution of Person Parameters

In CDM, for dichotomous responses and K attributes associated with a test, the number of all possible attribute patterns is 2^K . Hence, an unstructured latent model requires $2^K - 1$ structural parameters for 2^K possible latent classes or attribute patterns. In complex estimation scenarios, estimating $2^K - 1$ structural parameters would be inconvenient, and attributes in a test are correlated with each other (de la Torre & Douglas, 2004). To achieve a reduced parametric structure of the joint distribution, the response accuracy measurement model utilizes a higher-order latent structure (de la Torre & Douglas, 2004) to obtain attribute estimates. In the higher-order model, all attributes are assumed to be independent given θ (i.e., the higher-order ability estimates). The link function for the higher-order model used in the four-component CDM can be presented as

$$P_{nk} = P(\alpha_{nk} = 1 | \theta) = \frac{e^{(\theta_n - \gamma_k)}}{1 + e^{(\theta_n - \gamma_k)}}. \quad (5)$$

Here, P_{nk} is the likelihood of individual n possessing k -th attribute given a specific θ level, γ_k is the higher-order intercept for k -th attribute, and θ_n is the general (higher-order) ability for person n .

As mentioned earlier, we employ the hierarchical modeling framework to model the variance-covariance structure among person parameters from

measurement models of response accuracy, response time, similarity, and efficiency. We assume that person parameters of the four-component model follow a quadrivariate normal distribution:

$$\Lambda_n = \begin{pmatrix} \theta_n \\ \tau_n \\ s_n \\ \xi_n \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_\theta \\ \mu_\tau \\ \mu_s \\ \mu_\xi \end{pmatrix}, \Sigma_{\text{person}} \right); \quad (6)$$

$$\Sigma_{\text{person}} = \begin{pmatrix} \sigma_\theta^2 & \rho_{\theta\tau}\sigma_\theta\sigma_\tau & \rho_{\theta s}\sigma_\theta\sigma_s & \rho_{\theta\xi}\sigma_\theta\sigma_\xi \\ \rho_{\theta\tau}\sigma_\theta\sigma_\tau & \sigma_\tau^2 & \rho_{\tau s}\sigma_\tau\sigma_s & \rho_{\tau\xi}\sigma_\tau\sigma_\xi \\ \rho_{\theta s}\sigma_\theta\sigma_s & \rho_{\tau s}\sigma_\tau\sigma_s & \sigma_s^2 & \rho_{s\xi}\sigma_s\sigma_\xi \\ \rho_{\theta\xi}\sigma_\theta\sigma_\xi & \rho_{\tau\xi}\sigma_\tau\sigma_\xi & \rho_{s\xi}\sigma_s\sigma_\xi & \sigma_\xi^2 \end{pmatrix}.$$

In Equation 6, σ_θ^2 , σ_τ^2 , σ_s^2 , and σ_ξ^2 represent the variance of the ability parameter, response time parameter, response similarity parameter, and response efficiency parameter, respectively. Additionally, $\rho_{\theta\tau}$, $\rho_{\theta s}$, $\rho_{\theta\xi}$, $\rho_{\tau s}$, $\rho_{\tau\xi}$, and $\rho_{s\xi}$ represent the associations between ability and RT, ability and similarity, ability and efficiency, RT and similarity, RT and efficiency, and similarity and efficiency, respectively.

Identification of the Four-Component CDM

Several constraints need to be imposed for the scale identifiability. First, we assume $\mu_\theta = 0$ and $\sigma_\theta = 1$ (Bolsinova, Tijmstra, et al., 2017; Liang et al., 2023), which are used to establish the identifiability for the ability parameter (θ_n) and the higher-order parameter (γ_k). For the RT, similarity, and efficiency measurement models, this study sets $\mu_\tau = 0$ (van der Linden, 2007), $\mu_s = 0$, and $\mu_\xi = 0$. Similar restrictions are also used by several studies (please see: Fox & Mariani, 2016; Klein et al., 2009; Zhan et al., 2018). Additionally, to ensure monotonicity, all main effects of the LLM are assumed to be non-negative. Considering the conditional independence used by van der Linden (2007), the four-component model assumes that (a) $\log(T_{ni})|\tau_n$, $\text{logit}(S_{ni})|s_n$, $\text{logit}(E_{ni})|\xi_n$, and $\alpha_{nk}|\theta_n$ are all independent, (b) Responses (Y_{ni}) are conditionally independent for any given attribute (α_{nk}), and (c) Responses (Y_{ni}), $\log(T_{ni})$, $\text{logit}(S_{ni})$, and $\text{logit}(E_{ni})$ are conditionally independent given person parameters.

Parameter Estimation

The Markov Chain Monte Carlo (MCMC) method was used to estimate the parameters of the four-component CDM, denoted by Θ , which include incidental parameters

$$\Theta_{\text{incidental}} = [\theta_n, \tau_n, s_n, \xi_n, \alpha_{nk}]$$

and structural parameters

$$\Theta_{\text{structural}} = \left[\beta_i^{(T)}, \beta_i^{(S)}, \beta_i^{(E)}, \Gamma_k, \Lambda_k, \sigma_{Ti}^2, \sigma_{Si}^2, \sigma_{Ei}^2, \Sigma_{\text{person}} \right],$$

where, $\Gamma_k = [\gamma_1, \gamma_2, \dots, \gamma_k]$, $\Lambda_k = [\lambda_{0i}, \lambda_{1i}, \dots, \lambda_{Ki}]$; $n = 1, \dots, N$; $k = 1, 2, \dots, K$; $i = 1, 2, \dots, I$.

Under the assumptions of local independence, the likelihood function of the four-component model is specified as

$$L(Y, \log(T), \logit(S), \logit(E) | \Theta) = \prod_{n=1}^N \prod_{i=1}^I P(Y_{ni} | \alpha_n, \Lambda_k) f(\log(T_{ni}) | \beta_i^{(T)}, \tau_n, \sigma_{Ti}^2) \\ f(\logit(S_{ni}) | \beta_i^{(S)}, s_n, \sigma_{Si}^2) \times f(\logit(E_{ni}) | \beta_i^{(E)}, \xi_n, \sigma_{Ei}^2). \quad (7)$$

The joint posterior distribution can be written by

$$P(\Theta | Y, \log(T), \logit(S), \logit(E)) \propto L(Y, \log(T), \logit(S), \logit(E) | \Theta) \\ \times P(\alpha | \Gamma_k, \theta) P(\Gamma_k | \theta, \tau, s, \xi | \Sigma_{\text{person}}) P(\Sigma_{\text{person}}) \\ \times P(\Lambda_k) P(\beta_i^{(T)}) P(\beta_i^{(S)}) P(\beta_i^{(E)}) P(\sigma_{Ti}^2) P(\sigma_{Si}^2) P(\sigma_{Ei}^2). \quad (8)$$

To estimate the model parameters from the four-component model, we need to specify priors for all parameters and the variance–covariance matrix, that is, Σ_{person} . To specify prior for Σ_{person} , an inverse-Wishart prior is generally used, as it is a natural conjugate prior for the variance–covariance matrix (Barnard et al., 2000). However, the inverse-Wishart prior is not applicable for Σ_{person} since the variance of θ_n is assumed to be 1 to ensure identifiability (Zhan et al., 2018). Hence, to specify prior for Σ_{person} , we reparametrized Σ_{person} . Following Barnard et al. (2000), the variance–covariance matrix of the person parameters, Σ_{person} is factorized as

$$\Sigma_{\text{person}} = \mathcal{S} \mathcal{C} \mathcal{S}, \quad (9)$$

where $\mathcal{S}$ is a diagonal matrix comprised of the standard deviations of person parameters:

$$\mathcal{S} = \begin{pmatrix} \sigma_\theta & 0 & 0 & 0 \\ 0 & \sigma_\tau & 0 & 0 \\ 0 & 0 & \sigma_s & 0 \\ 0 & 0 & 0 & \sigma_\xi \end{pmatrix}. \quad (10)$$

$\mathcal{S}$ is the correlation matrix for person parameters.

$$\mathbf{C} = \begin{pmatrix} 1 & \rho_{\theta\tau} & \rho_{\theta\varsigma} & \rho_{\theta\xi} \\ \rho_{\tau\theta} & 1 & \rho_{\tau\varsigma} & \rho_{\tau\xi} \\ \rho_{\varsigma\theta} & \rho_{\varsigma\tau} & 1 & \rho_{\varsigma\xi} \\ \rho_{\xi\theta} & \rho_{\xi\tau} & \rho_{\xi\varsigma} & 1 \end{pmatrix}. \quad (11)$$

Furthermore, the correlation matrix, $\mathbf{C}$, can further be factorized using the Cholesky factorization (Sorensen & Vasissth, 2015), $\mathbf{C} = \mathbf{U}\mathbf{U}^*$. Here, $\mathbf{U}^*$ is an upper triangular matrix. As suggested by Nimble Development Team (2021), using the Lewandowski-Kurowicka-Joe (LKJ) distribution (Lewandowski et al., 2009) as a prior for $\mathbf{U}^*$ is computationally more efficient. LKJ distribution is a prior distribution specified for correlation matrices or their Cholesky factors (Rezaee et al., 2023). Hence, we specify prior distribution on $\mathbf{U}^*$ as $\mathbf{U}^* \sim \text{dlkj_corr_cholesky}(1.3)$. Conventionally, an inverse-gamma prior is chosen for variance components. However, for inverse-gamma (ϵ, ϵ) prior distribution, if $\epsilon \rightarrow 0$, it leads to an improper posterior density (Gelman, 2006). To avoid such issues, and as we employed weakly informative priors for variance components, the prior distributions of σ_τ , σ_ς , and σ_ξ are specified as $N(0, 5^2)$ truncated at above 0. Following Liao and Jiao (2020), the priors for LLM parameters are assigned as $\lambda_0 \sim N(-1.096, 2^2)I(\lambda_0 < 0)$, and $\lambda_k \sim N(0, 2^2)I(\lambda_k > 0)$. Furthermore, the priors for item parameters of the four-component model are specified as $\beta_i^{(T)} \sim N(0, 5^2)$, $\beta_i^{(S)} \sim N(0, 5^2)$, $\beta_i^{(E)} \sim N(0, 5^2)$, $\sigma_{Ti} \sim N(0, 5^2)I(\sigma_{Ti} > 0)$, $\sigma_{Si} \sim N(0, 5^2)I(\sigma_{Si} > 0)$, and $\sigma_{Ei} \sim N(0, 5^2)I(\sigma_{Ei} > 0)$. This study utilized Numerical Inference for Statistical Models using Bayesian and Likelihood Estimation (NIMBLE; de Valpine et al., 2017) in R (version 4.2.2; R Core Team, 2022) to estimate the four-component model parameters.¹

Application of Four-Component Model to PIAAC Data

In this section, we fit the four-component model to a set of data in PSTRE from the PIAAC Survey 2012 (OECD, 2012). To assess the implications of ignoring RT, similarity, and efficiency information in the CDM analysis, we further employed two existing models, that is, the two-component model with item response and response time, and the higher-order LLM ignoring all process information. The two-component model is a modified version of the Joint Response Time - Deterministic Input, noisy 'and' gate (JRT-DINA; Zhan et al., 2018), and Higher-Order Linear Logistic Model (HO-LLM; Maris, 1999) is a widely used CDM. In addition, we fit the three-component model with similarity (ignoring efficiency in the four-component model), and the three-component model with efficiency (ignoring similarity in the four-component model) to the PIAAC data to explore the impacts of similarity and efficiency separately. In general, all additional models considered in the empirical analysis are special cases of the four-component model.

TABLE 1.
Q-Matrix for PSTRE Module-Two of PIAAC 2012 Survey

Items	E-mail (α_1)	Web (α_2)	Spreadsheet (α_3)
Unit 19a	1	0	1
Unit 19b	1	0	1
Unit 07	0	1	0
Unit 02	1	1	0
Unit 16	1	0	0
Unit 11b	1	0	0
Unit 23	1	1	0

Note. 1 represents the attribute/skill is required to answer respective questions, 0 otherwise.
PIAAC = Program for the International Assessment of Adult Competencies; PSTRE = problem solving in technology-rich environments.

In the PIAAC data, the PSTRE items were divided into two modules, each with seven items. This study uses items from Module 2, among which three were dichotomous and four were polytomous items. For modeling purposes, all polytomous items were dichotomized based on response accuracy (i.e., item response is converted into binary categories, correct and incorrect, and the partial credit is categorized into incorrectness). Table 1 exhibits the Q-matrix, which involves three attributes, e-mail use (α_1), web use (α_2), and spreadsheet use (α_3), corresponding with the technical environment dimensions in the PSTRE framework (OECD, 2016).

The number of reference sequences for PSTRE Module 2 items may vary across items. Of the seven items in PSTRE Module 2, only item U23 is designed for a unique reference action sequence. The number of reference sequences varies from 2 to 18 among the remaining six items (He et al., 2021). In the presence of multiple reference sequences for an item, similarity is measured by assessing the deviation of a respondent's action sequence from the closest reference sequence. Efficiency is measured based on the number of redundant actions performed by the examinee compared to the closest reference sequence. A higher number of redundant actions indicates lower efficiency for a respondent. This study focused on 935 respondents from the U.S. sample who completed all items of PSTRE Module 2. Any response with missing data was ignored in this study.

Analysis

We initiated five Markov chains, each with 5,00,000 iterations with a thinning interval of 10 to estimate the four-component model (i.e., response accuracy, RT, similarity, and efficiency) and two three-component models (i.e., response accuracy, RT, and similarity or efficiency). The first 250,000 iterations

were considered as burn-ins. For the two-component model and HO-LLM, we achieved convergence using 30,000 iterations and 10,000 burn-ins. $\hat{R}$ was obtained for each parameter. Posterior predictive model checking (Gelman et al., 2014) was employed to assess whether each measurement model sufficiently fit the data for respective measurement aspects separately. In particular, this study generates the posterior predictive p -values (PPP) of discrepancy measures for assessing the deviation in observed and replicated values (Ulitzsch et al., 2020). A PPP value close to 0.50 indicates the absence of a systematic difference between the observed and the replicated values. The exploration of model-fit statistics for joint models has been limited (Zhan et al., 2018). Hence, we obtained separate model-data fit statistics for each measurement model in the four-component model. As the response accuracy measurement model is defined based on dichotomous response data, we used the sum of the squared Pearson residuals as the discrepancy measure (Almond et al., 2015; Yan et al., 2003; Zhan et al., 2018), which is expressed as

$$D(Y_{ni}; \alpha_{nli}^*) = \sum_{n=1}^N \sum_{i=1}^I \left(\frac{Y_{ni} - P(Y_{ni} = 1)}{\sqrt{P(Y_{ni} = 1)(1 - P(Y_{ni} = 1))}} \right)^2, \quad (12)$$

where $P(Y_{ni} = 1)$ is defined in Equation 1. For the RT, similarity, and efficiency, we employed the sum of the squared standardized error function of $\log(T_{ni})$, $\text{logit}(S_{ni})$, and $\text{logit}(E_{ni})$ (Fox & Marianti, 2017; Zhan et al., 2018) as the discrepancy measures. The sum of squared standardized error functions of $\log(T_{ni})$ is provided below:

$$D(\log(T_{ni}); \beta_i^{(T)}, \tau_n, \sigma_{Ti}) = \sum_{n=1}^N \sum_{i=1}^I \left(\frac{\log(T_{ni}) - (\tau_n - \beta_i^{(T)})}{\sigma_{Ti}} \right)^2. \quad (13)$$

$D(\text{logit}(S_{ni}); \beta_i^{(S)}, \varsigma_n, \sigma_{Si})$ and $D(\text{logit}(E_{ni}); \beta_i^{(E)}, \xi_n, \sigma_{Ei})$ are also obtained similar to Equation 13. In addition, we obtained the deviance information criterion (DIC; Spiegelhalter et al., 2002) for item response models. Here, $\text{DIC} = \overline{D(\Phi)} + p_e = \overline{D(\Phi)} + 0.5 \times \text{var}(D(\Phi))$ (Gelman, 2004; Shriner & Yi, 2009; Su & Yajima, 2015), where, $D(\Phi) = D(Y, \Phi) = -2\ln(P(Y|\Phi))$, Y represents the observed item response data, and Φ represents the vector of item parameters.

Results

Model Fit. The Gelman-Rubin statistic ($\hat{R}$) for all parameters for all models was less than 1.05, indicating that parameter estimation converged adequately (Gelman et al., 2014). In the four-component model, the PPP values for response accuracy, RT, similarity, and efficiency models were 0.662, 0.516, 0.515, and

0.514, respectively, indicating that the measurement models used in the four-component model fit the PIAAC data adequately. The item-wise PPP value for each measurement model in the four-component model is provided in Table S1 (available in the online version of this article). The item-wise PPP values for response accuracy measurement model ranges in [0.242, 0.436], and the item-wise PPP values for RT, similarity, and efficiency measurement models are close to 0.5, which are covered within the acceptable range of [0.05, 0.95] (Zhan et al., 2019 citing Gelman et al., 2014). PPP values for all other models indicated adequate model fit to the PIAAC data (see Table S2 in the online version of the journal). The model-data fit results suggest that the four-component model in conjunction with the Q-matrix considered in the study can fit the data adequately and provide meaningful diagnostic information. Furthermore, we obtained DICs for item response models employed in the study. The DIC values for the item response models are as follows: 6,574.08 for the four-component model, 6,571.47 for the three-component model with similarity, 8,886.51 for the three-component model with efficiency, 8,559.29 for the two-component model, and 8,695.12 for the HO-LLM. The DIC for the four-component model and the three-component model with similarity are very close, and much lower than that from the three-component model with efficiency, the two-component model and the HO-LLM.

Correlations Among the Process Components. Exploring the associations among PSTRE proficiencies, similarity, and efficiency, He et al. (2021) concluded that similarity and PSTRE proficiency are highly correlated, whereas efficiency is moderately correlated with PSTRE proficiency levels. We further examined item-wise correlation structure among RT, similarity, and efficiency, which may offer us initial insights into the correlation of corresponding latent variables. We first examined item-wise correlation among RT, similarity, and efficiency. As shown in Table S3 (available in the online version of this article), the similarity and efficiency parameters generally showed a negative correlation for all items except for U11b. In item U11b, the similarity and efficiency parameters resulted in a positive correlation .224, suggesting that when people show closer action sequences against reference sequences, they also intend to use more concise steps to reduce the redundant actions. It is also noted that the absolute value of correlation coefficients for items U02 and U16 is much higher than the other items, indicating that when solving these two items, it is more likely for respondents to use longer action sequences (i.e., redundant actions) in order to find the closer strategies compared with the reference sequences. We also noticed that the RT and efficiency parameters always showed a moderately negative correlation, suggesting the longer response time is more likely to be associated with a low efficiency (i.e., using more actions than necessary). Furthermore, the correlation between RT and similarity resulted in a moderate

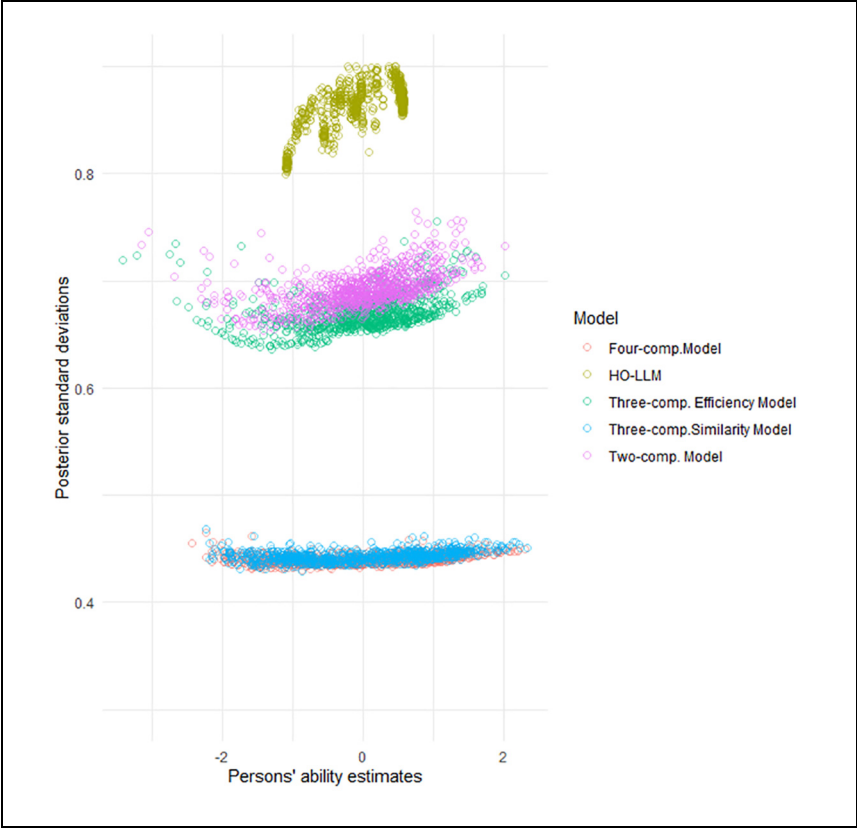


FIGURE 2. *Posterior standard deviations of ability estimates for different models.*

positive correlation in items U07 and U02. It suggested that respondents intended to spend more time on these two items to get closer strategies compared with the predefined optimal ones.

Comparison of Empirical Results. As demonstrated by Liang et al. (2023), we employed the posterior standard deviations (PSDs) to explore the accuracy of ability estimates. For all five models, employed in the empirical PIAAC data analysis, ability estimates (θ) and their corresponding PSDs (Bayesian standard error) are presented in Figure 2. Among all models, the four-component model mostly produced the lowest PSDs, followed by the three-component model with similarity. The lowest PSDs in the four-component model can be attributable to

the process data components considered in the model, which indicates that the four-component model can potentially use more information from the process data, yielding less uncertainty in parameter estimates (Liang et al., 2023). Comparing PSDs in the three-component model with similarity and the three-component model with efficiency, it can be concluded that similarity contributed more information (compared to efficiency) in the four-component model.

Parameter Estimates and Discussion. As shown in Table 2, the estimated correlation between the ability parameter and the RT parameter was 0.411, which indicated that ability is positively correlated with RT. This finding echoes the results from several previous studies (see Fox & Mariani, 2016; Meng et al., 2015; van der Linden, 2007). Test-takers who spent a longer time are more likely to answer PSTRE items correctly. Furthermore, the estimated correlation between latent ability and similarity parameters was as high as 0.990, which indicated that a person's latent ability is highly positively associated with the action sequences he/she employed.

The closer to the reference sequence, the higher the similarity score is achieved, which also echoes the findings in He et al. (2021). The correlation coefficient between ability and efficiency was $-.543$, suggesting individuals with higher abilities use more actions than the reference sequence while responding to PSTRE items. This finding reinforced the conclusion from Liang et al. (2023) that examinees with high scores were also found to take more actions to answer PSTRE items.

Table 3 presents the estimated item parameters from the four-component CDM. Of all the items, respondents had the highest average response time in item U02. As the population RT parameter (i.e., $\beta_i^{(T)}$) is estimated from a log-normal model, it cannot be compared directly with general RT measurements. However, considering all other factors as constant, the exponent of the time parameter (i.e., $\beta_i^{(T)}$) can be explained as the median time respondents take to answer specific items. For instance, the estimated value of the population RT parameter for item U02 is $\beta_{U02} = 1.26$; hence, the median time for responding item U02 is $e^{\beta_{U02}} = 3.53$ min. Furthermore, λ_0 is the logit (probability) of success if the examinee fails to master any attribute in the test. The λ_0 estimates indicated that, if the respondents do not possess any test attributes, they are less likely to answer an item correctly. For example, based on the λ_0 U02 estimate obtained in this study, respondents have an extremely low probability of answering item U02 correctly. On a similar note, Liao et al. (2019) identified item U02 as the most difficult item for the U.S. population. From the estimated values of the population similarity parameter ($\beta_i^{(S)}$), it can be inferred that respondents performed more similar actions (compared to the optimum reference action sequence) while responding to item U11b, and least similar actions while responding to item U02. From the estimated values of the efficiency measures

TABLE 2.
Estimated Variance–Covariance Matrix for the Four-Component Model

Variance–covariance parameter, Σ_{person}	Variance–covariance elements	Mean (SD)	Correlation	95% Credible interval	
				Lower	Upper
$\rho_{\theta\tau}\sigma_{\theta}\sigma_{\tau}$	Covariance of ability and RT	0.142 (0.016)	.411	0.356	0.462
$\rho_{\theta\varsigma}\sigma_{\theta}\sigma_{\varsigma}$	Covariance of ability and similarity	0.863 (0.024)	.990	0.987	0.992
$\rho_{\theta\zeta}\sigma_{\theta}\sigma_{\zeta}$	Covariance of ability and efficiency	−0.186 (0.020)	−.543	−0.586	−0.498
$\rho_{\tau\varsigma}\sigma_{\tau}\sigma_{\varsigma}$	Covariance of RT and similarity	0.103 (0.012)	.342	0.284	0.397
$\rho_{\tau\zeta}\sigma_{\tau}\sigma_{\zeta}$	Covariance of RT and efficiency	−0.111 (0.008)	−.940	−0.950	−0.927
$\rho_{\varsigma\zeta}\sigma_{\varsigma}\sigma_{\zeta}$	Covariance of similarity and efficiency	−0.146 (0.017)	−.489	−0.536	−0.438
σ_{τ}^2	Variance of RT	0.119 (0.007)			
σ_{ς}^2	Variance of similarity	0.759 (0.041)			
σ_{ζ}^2	Variance of efficiency	0.117 (0.012)			

Note. SD = standard deviation.

Items	Item parameters																														
	λ_0			λ_1			λ_2			λ_3			$\beta_i^{(7)}$			$\beta_i^{(S)}$			$\beta_i^{(E)}$			σ_{π}			σ_S			σ_{Ei}			
	M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		M	SD		
Unit 19a	-1.084	.438		1.871	.339					3.120	1.086		.657	.019	1.248	.070	-.422	.025	.466	.012	1.945	.047			.678	.017					
Unit 19b	-.999	.419		2.173	.332					3.120	1.099		1.182	.018	1.714	.082	-.070	.048	.447	.012	2.343	.056			1.430	.034					
Unit 07	-.429	.120					1.770	.265					.637	.019	1.874	.094	.713	.039	.461	.012	2.720	.064			1.153	.027					
Unit 02	-6.679	.921		4.387	.656		5.428	.822					1.263	.024	-.015	.033	-.701	.052	.666	.017	.518	.029			1.558	.037					
Unit 16	-.894	.218		2.793	.266								.826	.020	.776	.073	-1.788	.037	.491	.013	2.049	.050			1.081	.026					
Unit 11b	-1.278	.191		1.814	.221								.488	.021	2.219	.085	-.329	.065	.533	.014	2.469	.059			1.969	.046					
Unit 23	-1.769	.236		2.274	.309		.719	.270					.479	.020	1.332	.077	.729	.078	.498	.013	2.186	.052			2.373	.055					

Note. M = posterior mean; SD = standard deviation.

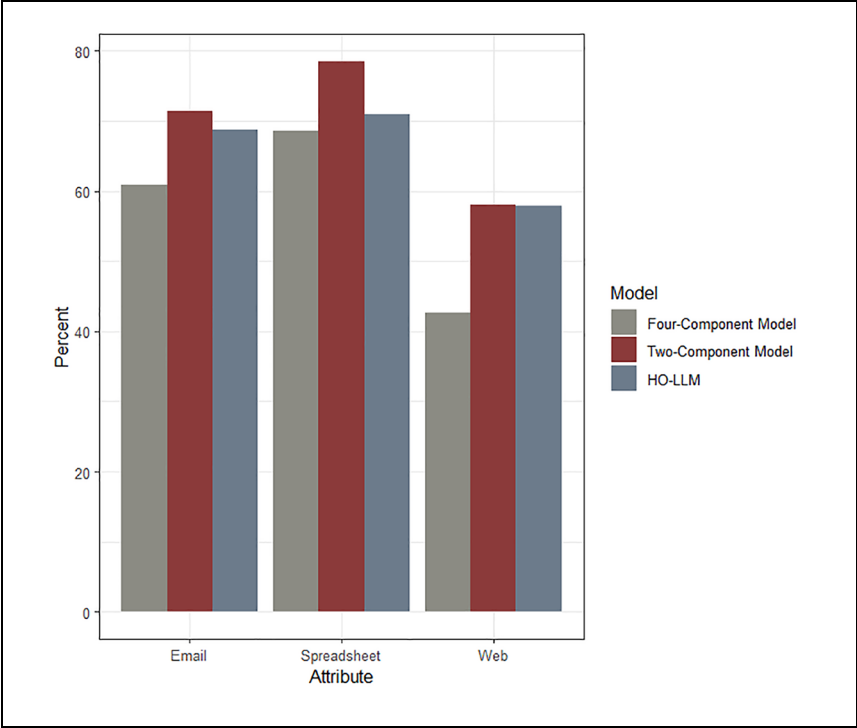


FIGURE 3. *Percentage of attribute mastery for the four-component model, two-component model and HO-LLM.*

(compared to LCS), it can be observed that, on average, test-takers were more efficient in responding to item U23, and least efficient in responding to item U16. We further explored the classification disagreement between the four-component model (with similarity, efficiency, and RT), two-component model (with response time), and the HO-LLM without any process data to examine how process data in the four-component model contribute to persons' classifications. Figure 3 presents a comparison of the percentage of mastery classifications for all attributes, as obtained from the four-component model, two-component model, and the HO-LLM. Based on the classification disagreement, we observed that the information from process data in the four-component model is contributing to a more conservative (i.e., lower) mastery classification as opposed to the two-component model and HO-LLM.

Based on the empirical analysis, we observed that the four-component model possesses certain advantages compared to the existing models considered in this

study. It provides additional information about test-takers' action sequences by offering estimates for similarity, and efficiency parameters for both individuals and items. Moreover, the four-component model provides estimates for covariances between the latent person parameters of ability, RT, similarity, and efficiency. This information is helpful for test developers to understand how a test-taker is attempting to solve each item. For instance, if test-takers' latent ability is negatively correlated with the corresponding latent similarity in a test, this could indicate potential aberrant behavior or problems with the test or its items. Furthermore, the efficiency measure may provide additional information about how carefully a test-taker is solving each item in a test. In addition, the four-component model demonstrates the lowest PSDs compared to the existing models, indicating lower estimation error for persons' ability estimates.

Simulation Study Based on Empirical Data Analysis

We compared the performance (in terms of parameter recovery and prediction accuracy) of the four-component model with models that either do not include process data or use reduced sets of process data, as considered in this study. To make such comparisons, a simulation study is designed. The primary goal of this simulation study is to examine (a) how well the four-component model recovers the parameters in the above PSTRE testing scenario and (b) the degree of impact after integrating information from RT, similarity, and efficiency in the CDM framework, on attribute classification accuracy (ACA).

Design

The estimated parameters were used as the true values for generating 30 sets of data for 1,000 respondents based on the estimates obtained in the four-component model. As considered in the empirical analysis, all five models (i.e., the four-component model, three-component model with similarity, three-component model with efficiency, two-component model, and HO-LLM) were fit to the simulated data. Following the empirical data analysis, similar MCMC settings were employed for estimating the parameters of the four-component model and its reduced three-component models. For estimating the two-component model and the HO-LLM, the MCMC approach was used with 30,000 iterations and 10,000 burn-ins. Bias and root mean square error (RMSE) are calculated for all five models' parameter estimates, and lastly, ACA and pattern-wise classification accuracy (PCA) were calculated and compared across all five models. Higher ACA and PCA values indicate better classification accuracy. The bias and RMSE were computed with Equations 14 and 15, respectively,

$$\text{bias} = \frac{\sum_{r=1}^R (\hat{P}_j - P_j)}{R}, \quad (14)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{r=1}^R (\hat{P}_j - P_j)^2}{R}}, \quad (15)$$

where P indicates the parameter of interest and R indicates the number of replications. A smaller value of bias and RMSE indicates better parameter recovery and more accurate estimation. To assess the classification accuracy for all attributes (i.e., e-mail, web, and spreadsheet), and individual attribute profiles, ACA and PCA were calculated. Considering $\hat{\alpha}_{nk}$ as the estimated attribute, and α_{nk} as the true attribute for individual n for attribute k , the ACA was calculated as

$$\text{ACA} = \frac{1}{R \times N \times K} \sum_{r=1}^R \sum_{n=1}^N \sum_{k=1}^K I(\hat{\alpha}_{nk}^{(r)} = \alpha_{nk}^{(r)}). \quad (16)$$

Also, assuming $\hat{\alpha}_n$ as the estimated attribute profile, and α_n as the true attribute profile for individual n , the PCA was calculated as

$$\text{PCA} = \frac{1}{R \times N} \sum_{r=1}^R \sum_{n=1}^N I(\hat{\alpha}_n^{(r)} = \alpha_n^{(r)}). \quad (17)$$

Results

The parameters of all five models converged adequately. The mean bias and RMSE of respondents' ability, speed, similarity, and efficiency parameters (i.e., θ , τ , ς , ξ) for all five models are presented in Table 4. The bias of persons' parameters for all five models was nearly zero, and the RMSE of persons' parameters was lowest for the four-component model, except for one instance (i.e., the RMSE of ability estimate from the four-component model is slightly higher than the RMSE of the three-component model with similarity).

Table 5 further demonstrates near-zero bias and low RMSE of variance-covariance parameters across all models. Comparing the bias and RMSE values presented in Table 6, the item response parameter estimates from the four-component model outperformed the estimates from the existing and reduced models, except for the estimate of λ_3 of the four-component model in items U19a and U19b. The bias of $\hat{\lambda}_{3,U19a}$ of the four-component model is a little higher than the bias of $\hat{\lambda}_{3,U19a}$ of the two-component model. Even though the bias of $\lambda_{3,U19a}$ of the four-component model is higher, the difference (i.e., 0.013) is not substantial.

The bias and RMSE of RT, similarity, and efficiency item parameters from the four-component model, three-component models, and the two-component model indicated good parameter recovery (see Tables S4 and S5 in the online version of the journal). The bias and RMSE for RT, similarity, and efficiency parameters were noted in a range of $[-0.02, 0.03]$ and $[0.02, 0.09]$,

TABLE 4.
Bias and RMSE of Persons' Ability, Speed, Similarity, and Efficiency Parameters

Models Parameters	Four-component model		Three-component model with similarity		Three-component model with efficiency		Two-component model		HO-LLM	
	Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE
θ	0.012	0.451	0.012	0.447	0.011	0.809	0.011	0.824	0.011	0.866
τ	0.001	0.155	0.001	0.906	0.001	0.914	0.001	0.164		
ς	0.010	0.386	0.010	0.460						
ξ	-0.001	0.166			-0.001	1.151				

Note. RMSE = root mean square error.

TABLE 5.
Bias and RMSE of Variance-Covariance Parameters of the Four-Component Model and Two-Component Model

Variance-Covariance parameter, Σ_{person}	Four-component model		Three-component with similarity model		Three-component with efficiency model		Two-component model	
	Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE
$\rho_{\theta\tau}\sigma_{\theta}\sigma_{\tau}$	-0.006	0.025	-0.002	0.023	0.007	0.028	0.007	0.028
$\rho_{\theta\varsigma}\sigma_{\theta}\sigma_{\varsigma}$	-0.060	0.068	-0.046	0.055				
$\rho_{\theta\xi}\sigma_{\theta}\sigma_{\xi}$	0.010	0.032			-0.004	0.038		
$\rho_{\tau\varsigma}\sigma_{\tau}\sigma_{\varsigma}$	-0.004	0.014	-0.001	0.013				
$\rho_{\tau\xi}\sigma_{\tau}\sigma_{\xi}$	0.002	0.009			0.001	0.008		
$\rho_{\varsigma\xi}\sigma_{\varsigma}\sigma_{\xi}$	0.009	0.021						
σ_{τ}^2	0.000	0.006	-0.001	0.006	-0.001	0.006	-0.001	0.005
σ_{ς}^2	-0.008	0.054	-0.010	0.054				
σ_{ξ}^2	0.005	0.014			0.003	0.014		

Note. RMSE= root mean square error.

TABLE 6.

Bias and RMSE of Item Parameters for All Five Models

Models	Item parameter	Items											
		Unit 19a		Unit 19b		Unit 07		Unit 02		Unit 16		Unit 11b	
		Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE	Bias	RMSE
Four-component model	λ_0	-0.068	0.220	-0.067	0.353	-0.087	0.159	1.426	1.486	0.086	0.210	0.041	0.203
	λ_1	0.057	0.243	0.039	0.227			-0.960	1.013	-0.017	0.253	-0.002	0.194
	λ_2					0.259	0.372	-1.240	1.288				
	λ_3	0.093	0.447	0.248	0.538								
Three-component model with similarity	λ_0	-0.069	0.226	-0.058	0.351	-0.086	0.156	1.386	1.498	0.083	0.209	0.041	0.203
	λ_1												
	λ_2	0.070	0.241	0.041	0.227	0.237	0.365	-0.940	1.013	-0.018	0.251	-0.003	0.192
	λ_3	0.077	0.445	0.218	0.535			-1.215	1.306				
Three-component model with efficiency	λ_0	-0.135	0.239	-0.150	0.354	-0.242	0.398	1.989	2.100	0.100	0.223	0.050	0.210
	λ_1												
	λ_2	0.125	0.241	0.098	0.240	0.738	0.905	-1.160	1.234	-0.040	0.261	-0.010	0.201
	λ_3	0.047	0.459	0.2186	0.534			-1.900	2.000				
Two-component model	λ_0	-0.132	0.259	-0.136	0.335	-0.239	0.374	2.095	2.147	0.108	0.221	0.048	0.208
	λ_1	0.123	0.241	0.098	0.230			-1.183	1.237	-0.048	0.256	-0.007	0.201
	λ_2					0.831	0.939	-1.992	2.027				
	λ_3	0.080	0.439	0.252	0.510								
HO-LLM	λ_0	-0.187	0.243	-0.156	0.300	-0.303	0.416	2.236	2.277	0.118	0.226	0.058	0.211
	λ_1	0.142	0.246	0.096	0.211			-1.226	1.278	-0.057	0.263	-0.017	0.200
	λ_2					1.031	1.115	-2.191	2.221				
	λ_3	0.127	0.376	0.223	0.390							-0.016	0.260

Note. Four-component model represents the joint model of item response, RT, similarity, and efficiency. Three-component model with similarity represents the joint model of item response, RT, and similarity. Three-component model with efficiency represents the joint model of item response, RT, and efficiency. Two-component model represents the joint model of item response and RT, and HO-LLM represents the higher-order linear logistic model. RMSE = root mean square error; RT = response time.

TABLE 7.
The Bias and RMSE of Higher-Order Parameter

Models	Higher-order parameter (γ)					
	γ_1		γ_2		γ_3	
	Bias	RMSE	Bias	RMSE	Bias	RMSE
Four-component model	.113	.179	.049	.204	-.108	.228
Three-component with similarity model	.109	.202	.042	.203	-.098	.227
Three-component with efficiency model	.119	.187	.070	.346	-.180	.259
Two-component model	.123	.187	.110	.354	-.160	.262
HO-LLM	.132	.192	.104	.308	-.203	.284

Note. RMSE = root mean square error.

respectively. Furthermore, the bias and RMSE of population RT parameter and standard deviation of RT residuals were consistently better or equivalent in the four-component model estimates compared with their peers. The bias and RMSE of the higher-order parameter are presented in Table 7. Consistent with our expectation, the bias and RMSE of the higher-order parameter (γ) in the four-component model outperformed the existing two-component and HO-LLM models.

Table 8 reports the ACA and PCA for all five models. The mean ACA and PCA for the four-component model resulted in 0.863 and 0.65, respectively, which are the highest compared with existing models, that is, the two-component model (ACA=0.84, PCA=0.60) and HO-LLM (ACA=0.84, PCA=0.59). The four-component model also outperformed the reduced three-component models in terms of ACA and PCA. After adding process data gradually to the baseline HO-LLM, the highest improvement in terms of ACA and PCA was observed in the four-component model, suggesting that RT, similarity, and efficiency data contributed to the improved ability estimation in the CDM framework. Drawing from the ACA and PCA results, it can be inferred that both similarity and efficiency contribute to the four-component model, with a higher degree of contribution from similarity.

Simulation Study Under Varied Simulation Conditions

We further continued a follow-up simulation study under varied conditions and evaluated the performance of the four-component model against two existing models, that is, the two-component model and the HO-LLM. The four-component model achieved higher ACA and PCA compared to its reduced

TABLE 8.
ACA and PCA for All Five Models

Models	ACA			PCA			
	α_1	α_2	α_3	Mean	SD	Min.	Max.
Four-component model	.894	.866	.828	0.863	0.007	0.650	0.691
Three-component model with similarity	.894	.866	.827	0.862	0.007	0.649	0.690
Three-component model with efficiency	.889	.823	.823	0.846	0.017	0.611	0.672
Two-component model	.890	.821	.823	0.845	0.017	0.609	0.672
HO-LLM	.889	.809	.822	0.84	0.016	0.598	0.651

Note. ACA = attribute classification accuracy; PCA = pattern-wise classification accuracy; SD = standard deviation; Min. = minimum; Max. = maximum.

three-component models in the simulation study based on empirical estimates. Therefore, we excluded three-component models from the simulation study under varied simulation conditions.

Design

We conducted the simulation study by exploring the effects from three factors and compared parameter recovery of three models (i.e., four-component model, two-component model and HO-LLM) under each varying condition. These three factors are: (a) number of items: $I = 15, 30$; (b) number of examinee or sample size: $N = 500, 1,000$; (c) lastly, the correlation between person parameters: high correlation (i.e., $\rho_{\theta\tau} = .7, \rho_{\theta s} = .7, \rho_{\theta\xi} = -.7, \rho_{\tau s} = .7, \rho_{\tau\xi} = -.7, \rho_{s\xi} = -.7$); low correlation (i.e., $\rho_{\theta\tau} = .3, \rho_{\theta s} = .3, \rho_{\theta\xi} = -.3, \rho_{\tau s} = .3, \rho_{\tau\xi} = -.3, \rho_{s\xi} = -.3$). Based on the empirical analysis, correlation between ability parameter (θ), person's speed parameter (τ) and similarity parameter (s) were assigned as negative correlation with persons' efficiency parameter (ξ). All other correlations were set as positive. The Q-matrix (with $I = 30$) considered for the simulation study was previously used by de la Torre and Chiu (2016) (see Table S6 in the online version of the journal). Furthermore, following Liang et al. (2023), the Q-matrix (with $I = 15$) was selected as a subset from the Q-matrix with 30 items, and identified with asterisks in Table S6 (available in the online version of this article). Both Q-matrices contain five attributes. In alignment with the empirical results, item parameters were generated using the truncated normal distributions, denoted by TN (mean, SD , min, max). In particular, $\lambda_0 \sim \text{TN}(-1.875, 2.15; -7, 0)$; $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5 \sim \text{TN}(2, 1; 0, 6)$; $\beta_i^{(T)} \sim \text{TN}(-0.791, 0.316; -1.5, 0)$; $\beta_i^{(S)} \sim \text{TN}(-1.305, 0.740; -3, 0.25)$; $\beta_i^{(E)} \sim \text{TN}(0.267, 0.868; -0.75, 2)$; $\sigma_T \sim \text{TN}(0.51, 0.07; 0, 1)$; $\sigma_S \sim \text{TN}(2.034, 0.716; 0, 3)$; $\sigma_E \sim \text{TN}(1.462, 0.569; 0, 1)$; $\gamma \sim U(-1, 1)$. The standard deviations for person speed parameter (σ_τ), similarity parameter (σ_s) and efficiency parameter (σ_ξ) were set to 0.5 for all simulation conditions. The person parameters were simulated from Equation 6 with zero-mean vector, and variance-covariance matrix, Σ_{person} (Σ_{person} varies based on simulation conditions). In this simulation design, the item parameters were kept fixed for a given condition, while person parameters varied across all replicated datasets. All three models were fit to all 30 sets of simulated data for each simulation condition and model parameters were estimated using the MCMC approach with 30,000 iterations and 10,000 burn-ins. Bias, RMSE, ACA, and PCA were calculated and compared across all three models.

Results

The parameters of all models converged adequately. Tables 9 and 10 present the recovery of person parameters by summarizing bias and RMSE for all simulation conditions. The mean bias and RMSE of all person parameters are close to zero, and all three models (i.e., four-component model, two-component model, and HO-LLM) reflect good abilities to recover person parameters. Furthermore, person parameters from the four-component model show consistently lower mean bias and mean RMSE compared to person parameters from the two-component model and HO-LLM. This result indicated that the four-component model was capable of recovering parameters more efficiently under high-correlation conditions (e.g., $\rho_{\theta\tau} = .7$, $\rho_{\theta\varsigma} = .7$, $\rho_{\theta\xi} = -.7$, $\rho_{\tau\varsigma} = .7$, $\rho_{\tau\xi} = -.7$, $\rho_{\varsigma\xi} = -.7$).

The classification accuracy of persons' attributes (ACA) and their attribute patterns (PCA) for all three models are presented in Table 11. It is observed that all the models have ACAs >0.90 for simulation conditions with 30 items, and the ACAs >0.84 for simulation conditions with 15 items. Furthermore, the correlation condition appears to have impact on the overall PCA of the four-component model. For instance, under the high correlation condition, it is observed that the PCAs of the four-component model were consistently slightly higher than the PCAs obtained for the two-component model and the HO-LLM, in most cases. However, with the low correlation condition, the four-component model did not exhibit any superior performance. If the correlation between persons' ability and process data parameters is low, using the four-component model might be inconvenient because the process data parameters may not provide meaningful additional information for estimating persons' attributes. Furthermore, for the simulation conditions with 15 items, the PCA of all three models varied for different sample sizes and correlation conditions, indicating that a higher number of items contributed more toward improved attribute/skill estimation. The summarized results regarding the recovery of item parameters, process data parameters, variance-covariances of person parameters, and higher-order parameters are presented in supplementary materials (see Tables S7 to S13 in the online version of the journal). Based on the results of the simulation study, the mean bias and RMSE of all item parameters (see Tables S7 to S10 in the online version of the journal) and persons' variance-covariance parameters (see Tables S11 and S12 in the online version of the journal), for all three models, are nearly zero under all simulation conditions, indicating that the models recovered the item and variance-covariance parameters adequately. In addition, the bias of all higher-order parameters (i.e., γ) for all attributes was close to zero for all three models, under all simulation conditions, and the corresponding RMSE was also low for these parameters (see Table S13 in the online version of the journal).

TABLE 9.
BIAS of Person Parameter Estimates Across All Replications

Correlation	<i>I</i>	<i>N</i>	Four-component model				Two-Component model		HO-LLM
			θ	τ	ς	ξ	θ	τ	θ
High	15	500	-.001	-.001	.001	.002	-.002	-.002	-.002
		1,000	.000	-.001	-.001	.004	.001	-.001	.001
	30	500	-.001	-.001	.001	.002	-.001	-.001	-.003
		1,000	.001	-.001	.001	.003	.001	-.001	.001
Low	15	500	-.003	.001	-.001	.003	-.003	.002	-.002
		1,000	.001	-.001	.001	.005	.001	-.001	.001
	30	500	-.001	.002	.001	.002	-.003	.003	-.002
		1,000	.001	-.001	.001	.005	.001	-.001	.001

Note. *N*=number of examinees; *I*=number of items.

TABLE 10.
RMSE of Person Parameter Estimates Across All Replications

Correlation	<i>I</i>	<i>N</i>	Four-component model				Two-component model		HO-LLM
			θ	τ	ς	ξ	θ	τ	θ
High	15	500	.580	.123	.217	.062	.639	.126	.781
		1,000	.572	.121	.214	.060	.634	.125	.780
	30	500	.564	.091	.220	.105	.613	.092	.747
		1,000	.560	.089	.216	.105	.611	.091	.749
Low	15	500	.746	.126	.245	.062	.762	.127	.782
		1,000	.742	.125	.242	.060	.760	.126	.781
	30	500	.706	.092	.246	.108	.719	.093	.738
		1,000	.715	.091	.243	.107	.730	.091	.750

Note. *N*=number of examinees; *I*=number of items.

In sum, the findings from the simulation study for varying simulation conditions suggested that the parameters of the four-component model can be well recovered using the MCMC estimation approach used in this study. Furthermore, the process data (i.e., RT, similarity, and efficiency) contribute additional information to the four-component model for more accurate ability/skill estimation. Lastly, under the high correlation conditions, the four-component model showed better PCAs.

TABLE 11.
ACA and PCA for All Three Models Under Varying Simulation Conditions

Correlation	N	I	Model	Attributes					ACA	PCA
				α_1	α_2	α_3	α_4	α_5		
High	500	15	Four-component	.869	.808	.855	.883	.824	0.848	0.457
			Two-component	.869	.805	.855	.883	.824	0.847	0.456
			HO-LLM	.867	.802	.852	.882	.816	0.844	0.448
			Four-component	.920	.948	.850	.904	.925	0.909	0.641
	500	30	Two-component	.919	.948	.850	.903	.925	0.909	0.639
			HO-LLM	.918	.947	.848	.904	.925	0.908	0.636
			Four-component	.872	.821	.868	.890	.831	0.856	0.479
			Two-component	.872	.820	.867	.889	.828	0.855	0.477
	1,000	15	HO-LLM	.870	.815	.864	.887	.827	0.853	0.471
			Four-component	.925	.946	.855	.909	.923	0.912	0.647
			Two-component	.925	.946	.854	.909	.923	0.912	0.646
			HO-LLM	.924	.946	.852	.908	.922	0.910	0.641
Low	500	15	Four-component	.867	.790	.853	.885	.817	0.842	0.442
			Two-component	.868	.797	.854	.883	.814	0.843	0.445
			HO-LLM	.867	.802	.852	.882	.816	0.844	0.448
			Four-component	.922	.944	.967	.928	.887	0.929	0.710
	500	30	Two-component	.921	.943	.968	.929	.887	0.929	0.710
			HO-LLM	.922	.944	.968	.928	.887	0.930	0.711
			Four-component	.870	.814	.866	.888	.827	0.853	0.470
			Two-component	.870	.814	.865	.888	.827	0.853	0.470
	1,000	15	HO-LLM	.870	.813	.864	.888	.828	0.852	0.470
			Four-component	.924	.947	.852	.908	.923	0.911	0.642
			Two-component	.924	.947	.852	.908	.923	0.911	0.643
			HO-LLM	.924	.947	.851	.908	.923	0.910	0.641

Note. N = number of examinees; I = number of items; ACA = attribute classification accuracy; PCA = pattern-wise classification accuracy.

Discussion and Conclusions

The utilization of process data demonstrated immense possibilities to enhance the estimation and interpretations of cognitive assessments. To extend the theoretical and practical aspects of augmenting multidimensional process data within the CDM framework, this study presented the four-component model. The four-component model integrated RT, action similarity and action efficiency with a conventional CDM to provide enhanced diagnostic inferences regarding test takers' ability, and response behavior regarding RT, similarity, and efficiency. The novelty of the four-component model lies within its augmentation of information from test-takers' action sequences (utilizing similarity and efficiency measures) in the conventional CDM framework along with RT information. Furthermore, the four-component model provides additional insights regarding the relationship between test takers' ability and their corresponding RT, similarity and efficiency while responding to a certain item. More importantly, the results from the four-component model confirmed that embedding information from RT, similarity and efficiency lowers the ability/skill estimation error of the conventional CDM. Along with the theoretical development of the four-component model, this study developed an estimation algorithm based on the Bayesian MCMC method to estimate the parameters. In sum, the four-component model is capable of estimating person parameters, item parameters, and parameters relevant to process data for individual test takers, and items. The empirical data analysis of the four-component model revealed that for technology-rich environment, persons' ability is positively associated with his/her response time, and action similarity. However, persons' ability was negatively associated with action efficiency indicating that test-takers in the higher ability group are more likely to employ longer action sequences to solve an interactive item, deviating from the optimum action sequence. The findings from the four-component model were aligned with the nature of decision-making process embedded with PIAAC 2012 PSTRE items, however, the findings regarding similarity and efficiency parameters could not be compared with any previous studies.

The simulation study, based on real-data analysis, demonstrated that the four-component model performed better in terms of parameter recovery and ability estimations compared to the base CDM (i.e., HO-LLM), the two-component model, and the reduced three-component models. Specifically, the four-component model exhibited lower bias and RMSE in most parameter estimates compared to the HO-LLM, the two-component model, and the reduced three-component models. Moreover, the ACA and PCA of the four-component model were consistently higher compared to the ACAs and PCAs of the existing models and the reduced three-component models. Additionally, the four-component model has a larger parametric structure and consequently, offers a

more comprehensive insight regarding cognitive assessments by obtaining additional information from RT, similarity, and efficiency data.

To further explore how the four-component model performs in terms of recovering the parameters and classifying attributes, a follow-up simulation study under varying simulation conditions was conducted. The results indicated that the four-component model was able to recover the model parameters adequately using the MCMC estimation procedure. In addition, it is observed that the four-component model was more efficient if the person's parameters were highly correlated. For instance, if the person's parameter (i.e., ability) is not associated with process data parameters, it might be inconvenient to use the four-component model. However, it is important to note that person's parameter is more likely to be highly associated with the person's response processes, and subsequently with relevant process data.

The important information from action sequences on assessment precision has aroused increasing attention recently. Qiao et al. (2023) adopted the Conway-Maxwell-Poisson distribution that accounts for different types of dispersion in action counts and incorporated it in the multiple group joint modeling of item responses, RT, and action counts in collaborative problem-solving tasks. Liang et al. (2023) proposed a JCDM-RRT model to incorporate item response, RT, and action counts of mouse clicks or drag traces into the CDM. These two studies both demonstrated significant improvements in parameter recovery and enhancement of classification accuracy with the addition of action-related process variables. These findings were further reinforced in the current study, where the four-component model resulted in a significant reduction in estimation errors in both empirical and simulation studies. Unlike the previous studies that only took the action counts into consideration, this present study incorporates the information of individual's action sequences from action similarity (how close to the reference sequence) and efficiency (how concise to the reference sequence), to describe the relationship between observed sequences against the reference sequences, which enriched the information from the mere count of actions.

Some limitations in the four-component model also merit discussion. The item response measurement model, that is, LLM considered in this study, can only accommodate binary response data. In this study, the item response was considered as a binary variable with two possible outcomes, that is, correct and incorrect. With polytomous or continuous item response, the four-component model may provide more detailed information (Minchen et al., 2017). Also, the study considered only binary attributes (mastery vs. non-mastery); however, using polytomous attributes may lead to better model accuracy (Chen & de la Torre, 2013; von Davier, 2008). The four-component model was developed based on certain assumptions, such as item response, RT, similarity, and efficiency being conditionally independent of a given person's parameters. Similar

assumptions were evident in other process data based joint modeling studies where IRT models or CDMs are considered for modeling item response data (please see Liang et al., 2023; van der Linden, 2007; Zhan et al., 2018). However, van der Linden (2007) identified the conditional independence of responses and RT data (given specific ability level) as counterintuitive as both responses and RT are nested within a specific set of test items and examinees. Further research is necessary to explore the conditional dependencies of RT, similarity, and efficiency parameters. In addition, a further evaluation of the estimation reliability of the four-component model would be valuable in the future study. Zhang et al. (2023) proposed a process-incorporated scoring method to enhance assessment precision of IRT-based latent proficiency estimates, and suggested using two measures (i.e., MSE and Kendall's τ) to compare the reliability of ability estimates. Though this scoring method was primarily focused on an IRT-based estimation procedure, an extension to CDM-based estimation would be possible to enhance the ACA in CDMs (Zhang et al., 2023). This will allow us to compare the prediction of latent attributes with different CDM models in future empirical studies. Furthermore, given the concerns about the unknown true abilities of examinees in Module 2, the ability estimates from Module 1 could serve as a benchmark to examine the ability estimates from the four-component model. However, since the predefined action sequences are not validated by the content experts for Module 1 items, we refrained from employing Module 1 data to obtain reference estimates from the four-component model. In future studies, we recommend further evaluation of the four-component model employing benchmark estimates.

The proposed four-component model is able to facilitate examiners and test developers to improve test quality and design, test timing, and overall cognitive assessment. It may lay foundations for some new research directions as well. The four-component model suggests that the similarity of actions (compared to the optimum action sequence) is highly correlated with the test-takers' ability. Hence, in future research, similarity measures can be explored as a validating measure for the estimated abilities. Furthermore, from the similarity estimates, it can be inferred how on average how the test takers are solving a problem. A low mean similarity score may indicate that the test takers are deviating from the optimum response sequence, indicating less clarity of the respective test item. Additionally, the interrelationships between response accuracy, RT, similarity, and efficiency may aid test designs for different environments. In sum, this study has successfully identified that similarity and efficiency have contributed to improving the classification accuracy of conventional CDM. Hence, this study creates the ground for integrating similarity and efficiency measures to improve cognitive assessment designs and item quality, detect test-takers' behavioral patterns, and enhance ability or skill estimation.

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Note

1. The NIMBLE code can be accessed here: <https://osf.io/aur4h/files/osfstorage/654e8e58b1613d09dcb7aca8>.

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